Welcome! We will begin shortly

Getting Started with Generative AI

Learning Outcomes

- Understand the role of Generative AI in business and its distinction from other AI paradigms.
- Explore different GenAI models and their applications across various industries.
- Implement GenAI to enhance automation, decision-making, and operational efficiency.

Guidelines

Listen only mode

Ask questions at the interest of the larger audience

Questions in the Q&A Box

Thank you

Kindly utilize the chat box for subject-relevant questions only to maximize your learnings from the session.

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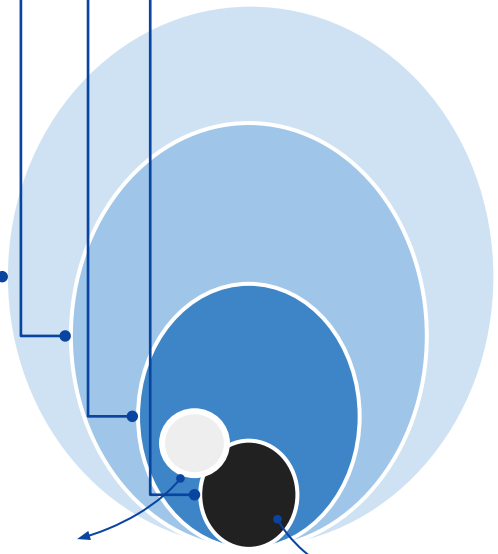
Key Terminologies in The World of Data

Artificial Intelligence *Machine that mimics human intelligence & behaviour*

Machine Learning *Machine that learns from the data w.r.t a particular task and performance measure*

Deep Learning *Subset of ML, learning happens via algorithms inspired by neural networks in the human brain*

Generative AI *Subset of DL, creating new data instances by learning the training data*



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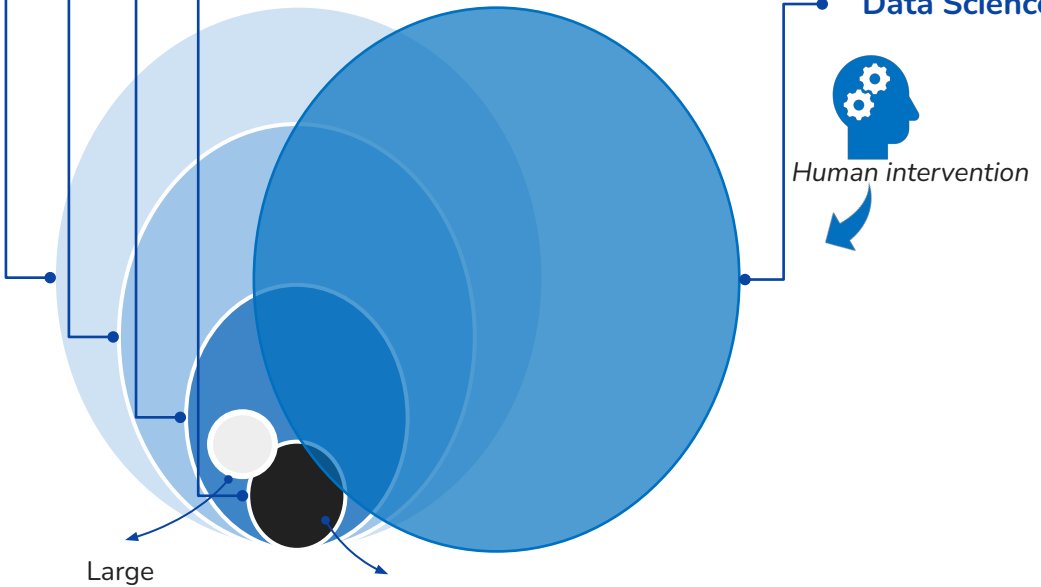
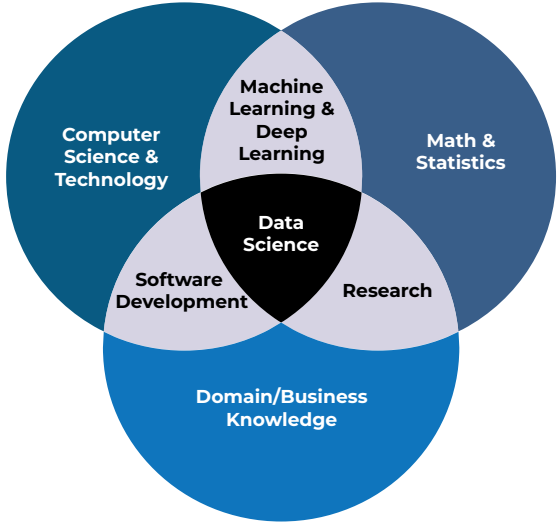
Data Science *Using AI to enable data driven decision making*





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Types of AI

Train

Model

Learns from relationship between **labels** and data

Could this be a dog?

Classify

YES

Discriminative AI

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Types of AI

Train

Model

Learns from relationship between **labels** and data

Could this be a dog?

Classify

YES

Discriminative AI

Train

Model

Learns from patterns in large amounts of unstructured data

+

Other images on the internet

Create an image of dog, having fun in a party, wearing a black tuxedo with wine in one hand

Generate

Generative AI

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Data Quiz

Which of the following is an example of a generative AI task?

A. Detecting whether an email is spam or not

B. Classifying images of animals into categories

C. Predicting whether a customer will buy a product or not

D. Creating a poem based on a topic given by the user

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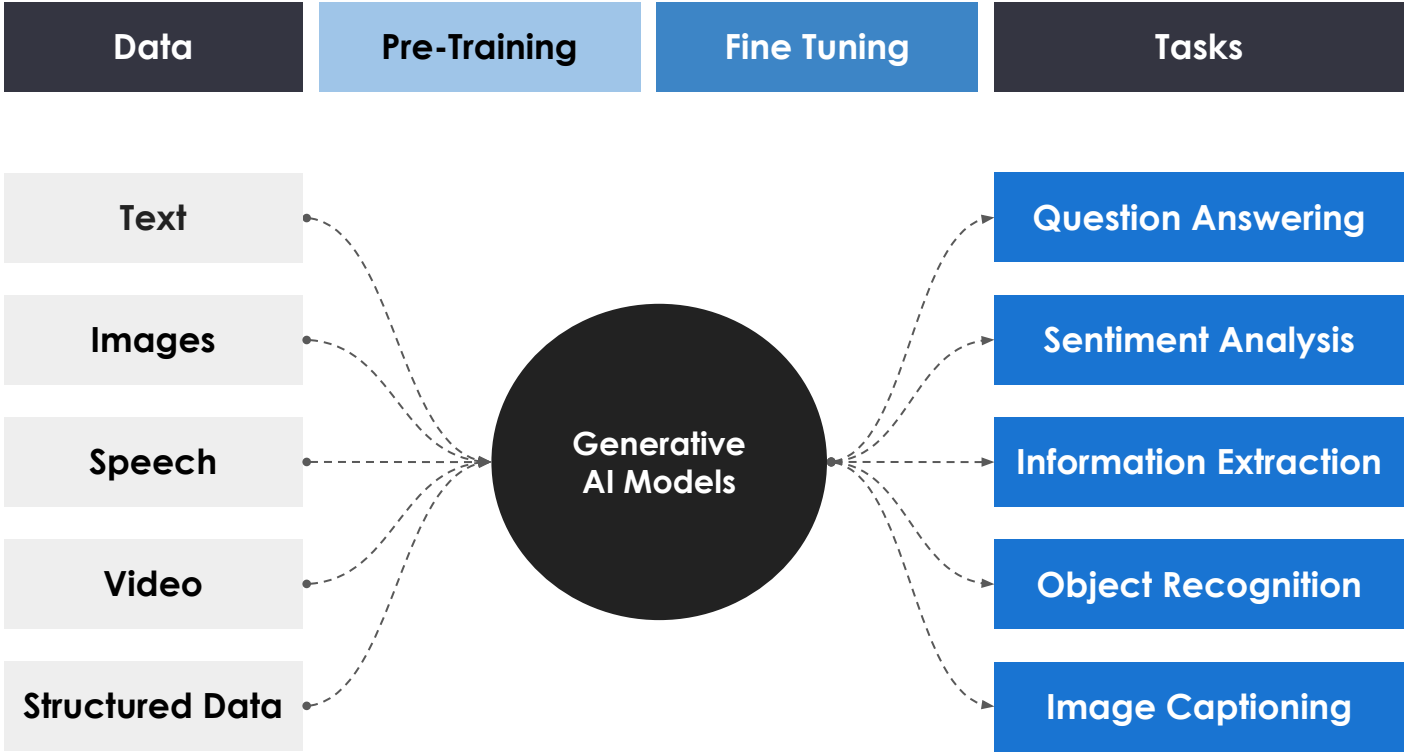
A Peek into Generative AI Models



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Large Language Models

Powerful AI models trained on massive amounts of data to learn the complex patterns and rules of human language, allowing them to perform a wide variety of tasks

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Large

Trained on **large amounts of data** and have **billions of trainable parameters**

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Language

Deals with text data (takes input in text and generates output in text)

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Language

Deals with text data (takes input in text and generates output in text)

Model

Predicts the next **token**

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Data Quiz

Which of the following best describes what a large language model does?

A. It guesses the next letter one by one until a full word is formed

B. It predicts the next word based on patterns learned from full sentences

C. It counts the number of letters in each word

D. It checks spelling and grammar only

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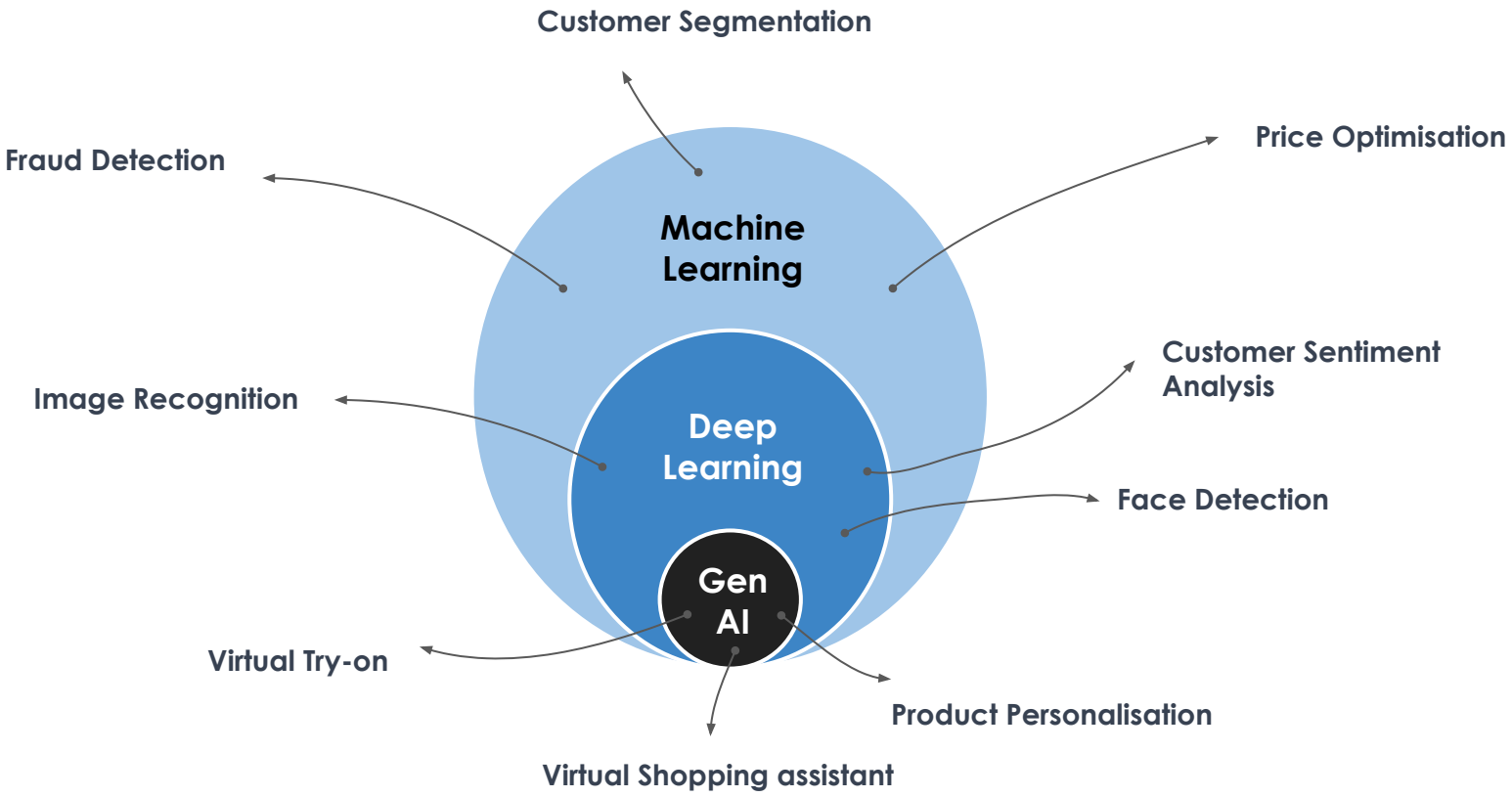
Business Problems Solved by Generative AI - Retail



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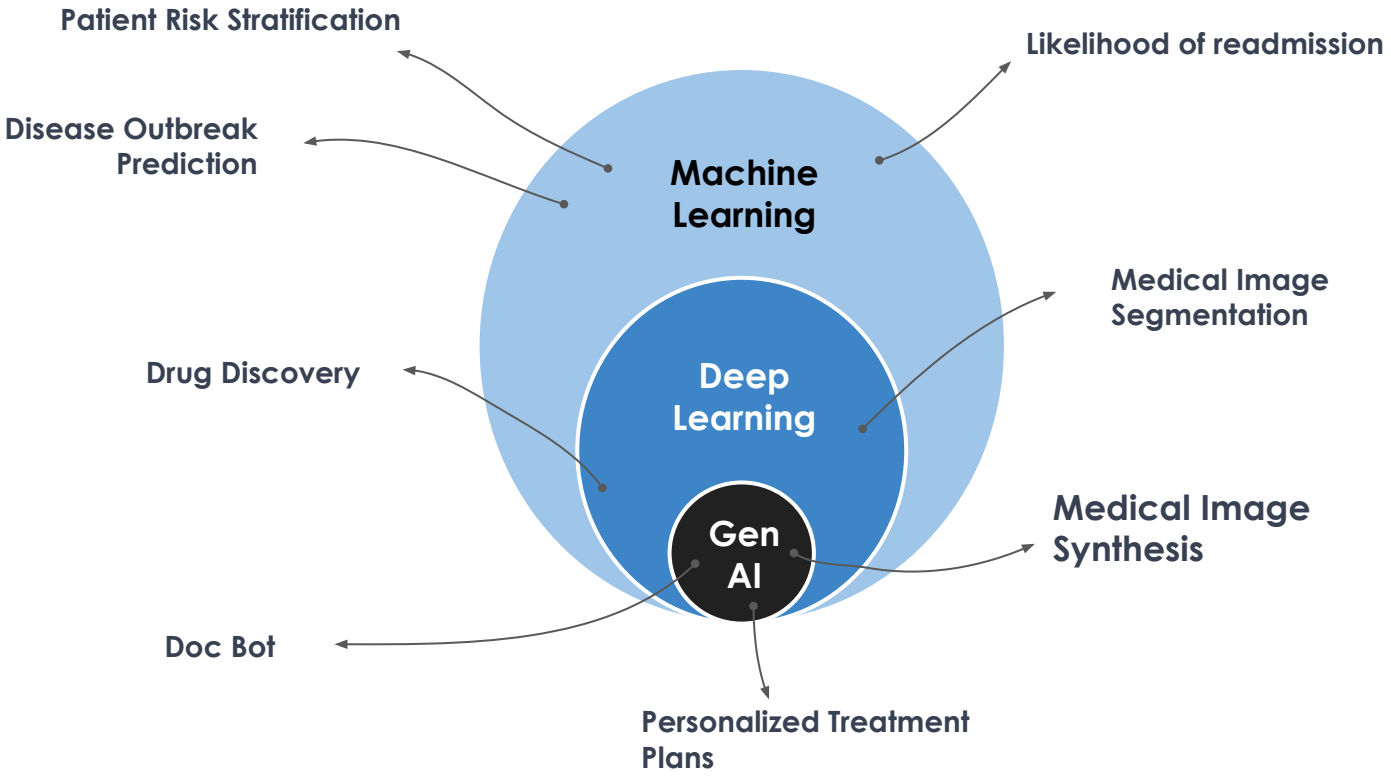
Business Problems Solved by Generative AI - Health care



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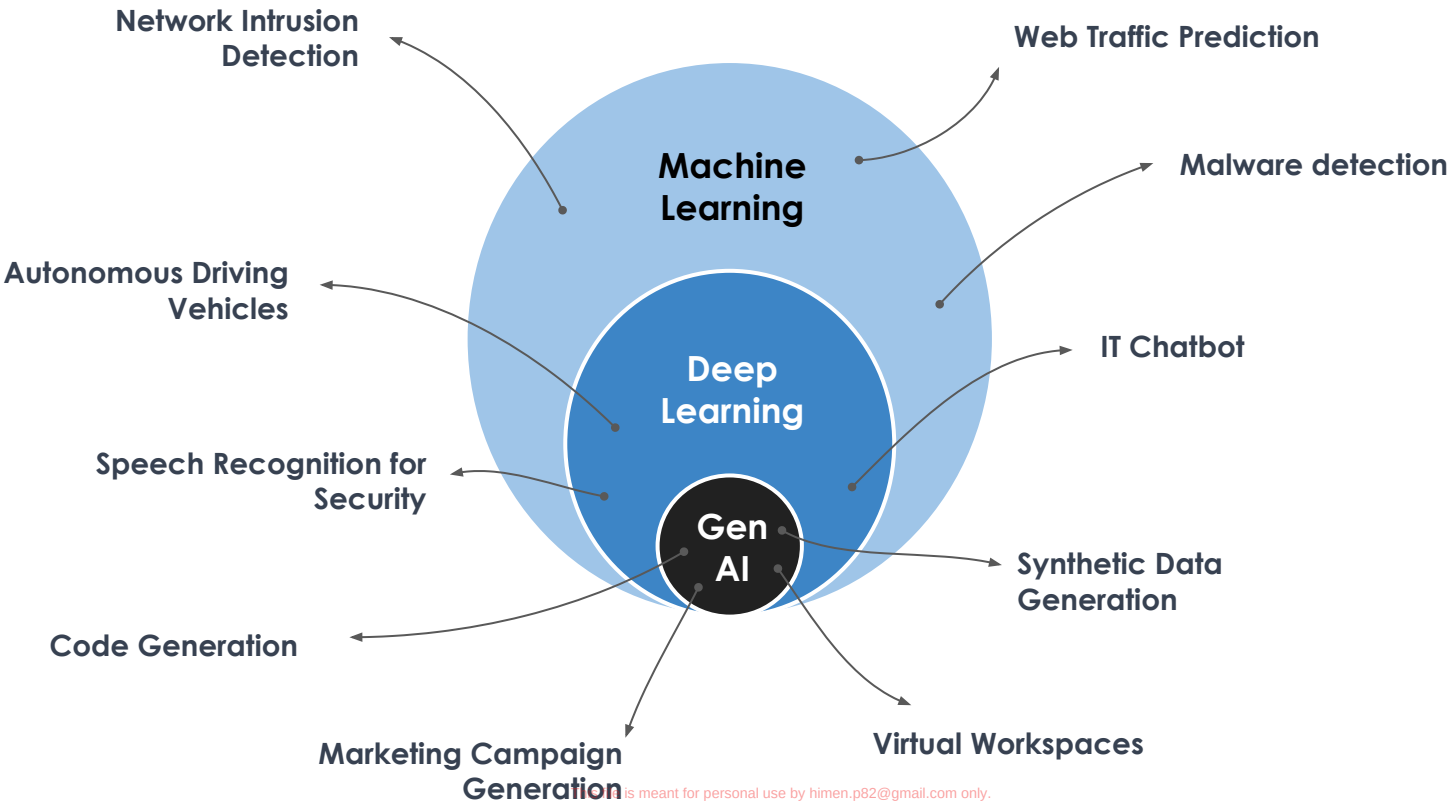
Business Problems Solved by Generative AI - Technology



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Data Quiz

Which of the following is LEAST LIKELY to be the output of a generative AI model?

A. A presentation for a lecture on data science based on provided instructions regarding content, audience, and session duration

B. Answers to questions regarding a popular tourist spot based on an images of the spot

C. A portrait of the Leaning Tower of Pisa with artistic features of a Leonardo da Vinci painting

D. Identifying whether a customer is likely to default on loan repayment based on credit history and other financial attributes

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Data Quiz

A presentation for a lecture on data science based on provided instructions regarding content, audience, and session duration

Generative AI models understand the relationship between instructions and presentation content by first learning from a collection of existing presentations and their instructions. They then 'decode' new instructions to generate relevant content.

Answers to questions regarding a popular tourist spot based on an images of the spot

The model first analyzes the visual features of the image (computer vision), then understands the question (natural language processing), and then generates a relevant answer based on its understanding of the image and the question

A portrait of the Leaning Tower of Pisa with artistic features of a Leonardo da Vinci painting

Generative AI models can transfer artistic styles between different image by identifying the underlying patterns. These style transfer abilities allow for the transformation of an image to adopt the artistic style of another image.

Identifying whether a customer is likely to default on loan repayment based on credit history and other financial attributes

Problems like the prediction of likelihood of loan repayment default are effectively dealt with by discriminative AI algorithms. They are generally more robust in these type of tasks as they learn the decision boundaries directly.

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Limitation of Large Language Models

I want to book a flight from Seattle to Portland for a meeting on June 18. I'm open to any time. What are the least expensive options available?

→ **Large Language Model** → ?

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Limitation of Large Language Models

I want to book a flight from Seattle to Portland for a meeting on June 18. I'm open to any time. What are the least expensive options available?

→ **Large Language Model** →

I don't have the ability to search for real-time flight information or book flights directly. However, I can suggest some tips on how to find the least expensive options for your flight from Seattle to Portland on June 18:

1. Flight comparison websites like Google Flights allow you to compare prices from different airlines. Set your departure city (Seattle) and arrival city (Portland) along with the date.

2. Then, from the available options, choose the flight with the lowest fare.

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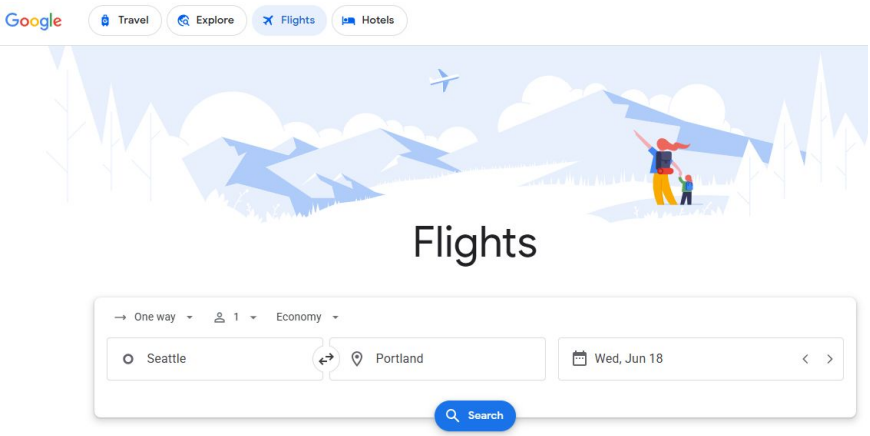
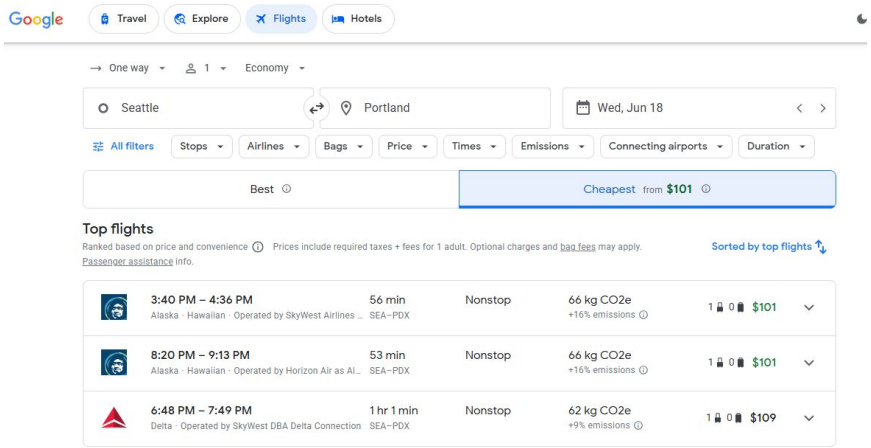
Limitation of Large Language Models

Searching Flights



Finding Cheapest Flight



- An LLM **CANNOT open a browser and search** for the cheapest flights - it **only takes text as input** and **produces text as output**

- But LLMs are **good at providing steps to search** for the cheapest flight

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Introduction to Tools

In the case of LLMs, **tools** are external capabilities that the model can use to perform tasks beyond text generation, such as retrieving live information, running code, or analyzing documents.

List of flights from Seattle to Portland on June 18

→

Google Search Tool

→

Alaska | 3:40 PM - 4:36 PM | 56 m | Nonstop | $101
Alaska | 8:20 PM - 9:13 PM | 53 m | Nonstop | $121
Delta | 6:48 PM - 7:49 PM | 1h 1m | Nonstop | $109
Alaska | 12:59 PM - 1:47 PM | 48 m | Nonstop | $131
Southwest | 8:50 AM - 5:25 PM | 8h 35 m | 1 stop (LAS) | $152
Delta | 4:05 PM - 5:10 PM | 1h 5 m | Nonstop | $189

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→ **Minimum Fare Tool** →

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If an **LLM** has **access to these tools**, it can easily **answer our original question**.

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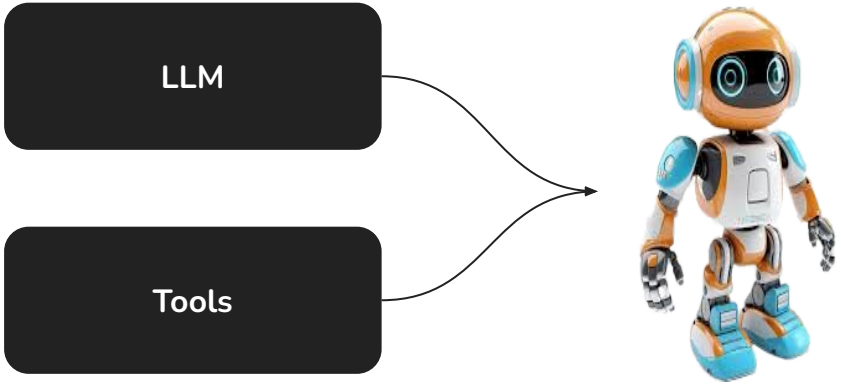
Introduction to Agent

An **AI agent** is a system that combines an LLM with tools while managing both the selection and use of the right tools to achieve the best results in answering a query.



AI Agent

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Agent in Action!

I want to book a flight from Seattle to Portland for a meeting on June 18. I'm open to any time. What are the least expensive options available?

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First, let me use the Google Search tool to get the flight details

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I want to book a flight from Seattle to Portland for a meeting on June 18. I'm open to any time. What are the least expensive options available?

Great! I got the flight details along with the fares and other information. To find the cheapest flight, I can use the Minimum Fare Tool

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Agent in Action!

I want to book a flight from Seattle to Portland for a meeting on June 18. I'm open to any time. What are the least expensive options available?

Great! Now I've also found the cheapest flight. Let me send the final result to the LLM to get the final output.

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Agent in Action!

I want to book a flight from Seattle to Portland for a meeting on June 18. I'm open to any time. What are the least expensive options available?

Great! Now I've also found the cheapest flight. Let me send the final result to the LLM to get the final output.

The least expensive flight from Seattle to Portland on June 18 is with Alaska: 3:40 PM - 4:36 PM, 56m, Nonstop, priced at $101.

LLM

Alaska | 3:40 PM - 4:36 PM | 56 m | Nonstop | $101

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Agent in Action!

Stopping the process now!

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Interacting with Generative AI Models

Hands-on: Exploring Generative AI Tools

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Thank you!

We'd love to hear your feedback!
Please share your feedback for the session

Wish you all the very best!

Please feel free to raise a Support Request through Olympus in case of any queries

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